

THE CENTRAL-CHARGE THRESHOLD FOR WALL-CROSSING IN TWO-VARIABLE OPEN FJRW THEORY

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ABSTRACT. Gross, Kelly and Tessler constructed a genus-zero open enumerative theory for the Landau–Ginzburg models $(x^r + y^s, \mu_r \times \mu_s)$, in which the open FJRW invariants depend on a choice of boundary conditions and jump across walls organized by a wall-crossing group. We determine exactly when this dependence occurs. Primary wall-crossing is governed by the central charge: it is absent precisely when $\hat{c}_W < 1$, that is, for the simple singularities A_{n-1} , D_4 , E_6 , E_8 , so the open theory detects the classical ADE threshold. The minimal number of interior insertions producing a wall obeys a uniform trichotomy; its smallest value, two insertions, occurs first for $x^4 + y^4$. A single descendent insertion produces wall-crossing for every model. Each wall-crossing generator is the Hamiltonian vector field, with respect to $dx \wedge dy$, of the boundary monomial of its critical disk, the monomial recording its boundary marked points; this exhibits the wall-crossing Lie algebra as a graded Hamiltonian subalgebra with tropical-vertex structure constants. Critical graphs of vanishing Euler weight (among them the unique first-wall graph of each parabolic pair) generate the torus direction excluded from the pro-nilpotent algebra, and it is there that the modularity of the simply-elliptic singularities first registers in the open theory. The threshold statement agrees with, and for the Fermat pairs gives an elementary proof of, a result established in greater generality in Maher’s thesis, of which we learned after this note was drafted. The classification of first walls and the Hamiltonian normal form are the starting point of the scattering-diagram construction of a companion paper. All computations are exact and are reproduced by code accompanying the note.

1. INTRODUCTION

Background. For a nondegenerate quasi-homogeneous polynomial W and a group G of diagonal symmetries, FJRW theory [FJR13] assigns to the Landau–Ginzburg model (W, G) a cohomological field theory: a genus-zero A -model built from moduli of W -spin curves and the Euler class of a Witten bundle. In their monograph [GKT22], Gross, Kelly and Tessler construct the *open* analogue for the two-variable Fermat model

$$W = x^r + y^s, \quad G = \mu_r \times \mu_s, \quad r, s \geq 2,$$

replacing closed W -spin curves by W -spin disks. The resulting open FJRW invariants are integrals of multisections of a descendent Witten bundle over moduli spaces that are real orbifolds with corners, and their central output is an open mirror theorem: a generating function of the open invariants is a versal deformation of W whose oscillatory period integrals recover the closed extended FJRW invariants, with the deformation parameters already flat [GKT22, Theorem 0.3].

A structural feature distinguishes the open theory from its closed counterpart and from the open r -spin theory of Buryak–Clader–Tessler [BCT24, BCT22]: the invariants are *not*

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canonical. Boundary conditions for the Witten bundle over the corners of moduli space are constrained, but not determined, by the positivity requirements of [GKT22]; the residual freedom makes the open invariants jump as the boundary data cross certain walls. Gross, Kelly and Tessler organize these jumps into a pro-nilpotent Landau–Ginzburg wall-crossing group $G^{r,s}$ acting on the set of possible systems of invariants, and prove that this action is faithful and transitive [GKT22, Theorems 0.7 and 4.26]: the systems of open invariants form a single torsor, while the period integrals – hence the closed theory – are constant along it.

The source of every wall is a *critical graph*: a stable W -spin graph Λ whose Witten bundle has rank one above the dimension of its moduli space, $\text{rank } E_\Lambda(\mathbf{d}) = \dim \overline{\mathcal{M}}_\Lambda^W + 1$ [GKT22, Definition 2.32]. Such a graph indexes a one-parameter family of codimension-one boundary strata through which zeros of the multisection can escape. The wall-crossing Lie algebra $\mathfrak{g}^{r,s}$ is a direct sum of one-dimensional pieces, one for each critical graph [GKT22, Definition 4.22]. Consequently the wall-crossing group is trivial – and the invariants are canonical – precisely when no critical graph exists.

The prediction. Whether critical graphs exist depends on (r, s) and on whether one allows descendents. In the descendent-free (*primary*) theory, [GKT22, Remark 0.9] predicts that wall-crossing is absent exactly for the simple singularities, and asserts this for an explicit list of pairs. As stated, that remark contains a discrepancy between its list and its naming, which we record and correct below. The prediction itself – that the primary open FJRW invariants are canonical if and only if W is a simple singularity – is left open in [GKT22] and is reiterated as such in the survey [GKT26].

Results. We prove the prediction and recast it as a clean invariant-theoretic threshold. Recall that the *central charge* of $W = x^r + y^s$, whose variables carry weights $q_x = 1/r$ and $q_y = 1/s$, is

$$(1) \quad \hat{c}_W = (1 - 2q_x) + (1 - 2q_y) = 2 - \frac{2}{r} - \frac{2}{s}.$$

Theorem 1.1 (Central-charge criterion). *Let $W = x^r + y^s$ with $r, s \geq 2$ and central charge $\hat{c}_W$ as in (1). The following are equivalent.*

- (i) $\hat{c}_W < 1$.
 - (ii) $\frac{1}{r} + \frac{1}{s} > \frac{1}{2}$.
 - (iii) W is a simple singularity: up to interchange of x and y ,
- $$(r, s) \in \{(n, 2) : n \geq 2\} \cup \{(3, 3), (3, 4), (3, 5)\},$$

of type A_{n-1} , D_4 , E_6 , E_8 respectively.

- (iv) For every finite marking set $I \subseteq \mathbb{N}$, the genus-zero primary open FJRW invariants of $(x^r + y^s, \mu_r \times \mu_s)$ are independent of the choice of symmetric canonical boundary conditions; equivalently, the descendent-free part of the wall-crossing group $G_{AI, \text{sym}}^{r,s}$ of [GKT22] (marginally enlarged as in Definition 1.2) acts trivially.

When these fail, that is when $\hat{c}_W \geq 1$, there is a finite marking set I for which the primary invariants genuinely depend on the boundary conditions.

Definition 1.2 (Euler weight; the marginally enlarged group). For interior data $(J, \mathbf{d})$ set $w(J, \mathbf{d}) = m(J, \mathbf{d}) - rs$, the *Euler weight*, where $m(J, \mathbf{d})$ is the mass of [GKT22, Notation 2.29] (its formula is recalled in Proposition 5.3); by the critical condition of [GKT22, Proposition 2.37], a critical graph with data $(J, \mathbf{d})$ and index (k_1, k_2) (the indexing of (6) below: the graph has k_1+1 and k_2+1 boundary marked points of the two types) satisfies

$sk_1 + rk_2 = w(J, \mathbf{d})$, so vanishing Euler weight means index $(0, 0)$. Such a *marginal* critical graph has generator the torus field $x\partial_x - y\partial_y$, the one direction that [GKT22, Definition 4.22] excludes for nilpotency of the filtration. The *marginally enlarged* wall-crossing Lie algebra and group are

$$\widetilde{\mathfrak{g}}^{r,s} = \mathfrak{g}^{r,s} \oplus \bigoplus_{w(J,\mathbf{d})=0, r(J)=s(J)=0} \mathbb{Q} u_{J,\mathbf{d}} (x\partial_x - y\partial_y), \quad \widetilde{G}^{r,s} = \exp(\widetilde{\mathfrak{g}}^{r,s}),$$

with $u_{J,\mathbf{d}} = \prod_{i \in J} t_{a_i, b_i, d_i}$ the deformation monomial; the exponential converges adically because every summand carries its monomial. The enlargement is forced on us: the proofs of [GKT22, Theorem 5.8 and Corollary 5.13] use exactly these summands (their jump sets impose only the critical condition, which admits the index $(0, 0)$), and their action realizes the explicit jumping of [GKT26, Example 4.9]; for the elliptic cases this is the group appearing in [Mah24, §5.2.2]. Faithfulness and transitivity extend: the enlarged group preserves the set of systems of open invariants, acts faithfully and transitively on it, and contains every jump automorphism of [GKT22, Theorem 5.8] ([Wue26, Lemma 2.4], whose proof runs through the realizability step (5.51)–(5.53) of [GKT22]). All statements below are over the enlarged group.

The equivalence (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) is elementary and classical; the content is (iv). The threshold $\hat{c}_W = 1$ is exactly the boundary between the simple singularities ($\hat{c}_W < 1$) and the parabolic (simple-elliptic, $\hat{c}_W = 1$) and hyperbolic ($\hat{c}_W > 1$) regimes. Thus the presence of primary wall-crossing detects, through open enumerative geometry, the classical ADE threshold.

Beyond the dichotomy, the same count determines *when* wall-crossing first appears. For an interior insertion the relevant twists lie in the range $0 \leq a \leq r - 2$, $0 \leq b \leq s - 2$; write m for the number of interior insertions (the mass $m(J, \mathbf{d})$, always written with its arguments, is a different quantity).

Theorem 1.3 (First wall). *Let $m_*(r, s)$ be the least number of primary interior insertions admitting a critical graph. Then*

$$m_*(r, s) = \begin{cases} \infty, & \frac{1}{r} + \frac{1}{s} > \frac{1}{2} \quad (\text{simple}), \\ 2, & \min(r, s) \geq 4, \\ 3, & \min(r, s) = 3 \text{ and } \max(r, s) \geq 6. \end{cases}$$

The case $m_* = 2$ is first realized by $W = x^4 + y^4$ with two interior insertions of twist $(2, 2)$, the example of [GKT26, §6].

Finally, descendents change the picture entirely: they generate walls in every case, including the simple singularities, as [GKT22, Remark 0.9] anticipates. The first such wall is universal.

Proposition 1.4 (First descendent wall). *For every (r, s) with $r, s \geq 2$, a single interior insertion carrying one descendent is a critical graph. Hence the open FJRW invariants with descendents exhibit wall-crossing for every two-variable Fermat model, simple or not.*

Where walls do occur, their generators have a uniform Hamiltonian description. Gross, Kelly and Tessler write the wall-crossing Lie algebra in terms of the vector fields $X_{a,b} = x^a y^b ((b+1)x\partial_x - (a+1)y\partial_y)$ [GKT22, eqn. (4.18)] and observe that they preserve $dx \wedge dy$ [GKT22, p. 85]. We name the Hamiltonian and read off the consequence.

Proposition 1.5 (Hamiltonian wall generators). *For $\omega = dx \wedge dy$ and $H_{a,b} = x^{a+1}y^{b+1}$, the wall generator $X_{a,b}$ is the Hamiltonian vector field of $H_{a,b}$; in particular it is divergence-free.*

The assignment $H_{a,b} \mapsto X_{a,b}$ is an isomorphism of graded Lie algebras from the Poisson algebra on the monomials $x^p y^q$, $p, q \geq 1$, onto its image, under which the bracket is the tropical-vertex determinant

$$[X_{a,b}, X_{c,d}] = ((b+1)(c+1) - (a+1)(d+1)) X_{a+c, b+d}.$$

It restricts to an isomorphism onto the wall-crossing Lie algebra $\mathfrak{g}^{r,s}$ of [GKT22], which is thereby the graded Hamiltonian subalgebra spanned by the critical Hamiltonians $\{H_\Lambda = x^{k_1+1} y^{k_2+1}\}$, where the critical graph Λ has k_1+1 and k_2+1 boundary marked points of the two types, so that H_Λ is precisely the boundary monomial of the critical disk. The (s, r) -weighted degree of H_Λ equals $w(\Lambda) + r + s$, with w the Euler weight of Definition 1.2; marginal graphs ($H_\Lambda = xy$) contribute the adjoined torus direction of Definition 1.2. This makes the tropical-vertex analogy of [GKT22, Remarks 0.8 and 4.19] an explicit map.

Proposition 1.5 completes the program the criterion was part of: an ADE/central-charge classification of when walls exist, a determination of the first wall, and a Hamiltonian normal form for the wall generators. The one structure it does *not* supply — a scattering diagram whose rays are the critical graphs — is constructed in the companion paper [Wue26], for which the census organized here is the starting point; see §5 and the closing section.

Method. Nothing here rebuilds the machinery of [GKT22]. Theorem 1.1 follows by combining three of its results: the closed-form count of critical graphs [GKT22, Notation 2.38], the identification of the wall-crossing Lie algebra with the span of the critical graphs [GKT22, Definition 4.22], and the faithfulness of the wall-crossing action [GKT22, Theorem 4.26]. The arithmetic core is a single inequality, Lemma 3.1, bounding the graph count by $m \hat{c}_W$. All numerical claims (the ADE list, the trichotomy $m_*(r, s)$, the first-wall census with its marginal/genuine split, and the universal descendent wall) are proved in §§3–6 and independently confirmed by the accompanying program `src/census.py`, which transcribes Notation 2.38 verbatim and checks these statements over a grid of (r, s) using only integer and rational arithmetic (§7).

Relation to prior work. The uniqueness of the primitive form for simple singularities is classical [Sai83, NY98], and is the closed-mirror shadow of the statement that these are the potentials without wall-crossing. On the open side, Basalaev–Buryak constructed open Saito potentials for the A - and D -series by solving the open WDVV equations [BB21]; the present criterion explains, uniformly, why the primary open theory is unambiguous on the whole ADE locus, including the E -cases, for which no closed-form potential has yet been computed. The wall-crossing group of [GKT22] is drawn, “in spirit,” by analogy with the tropical vertex group of Kontsevich–Soibelman [KS06] and its role in Fano mirror symmetry [Gro10] ([GKT22, Remark 0.8]); separately, a tropical version of open FJRW theory is raised as an open problem in [GKT26, §6], there through tropical moduli. The census of critical graphs organized here is a prerequisite for a scattering-diagram form combining the two, carried out in [Wue26]. After this note was drafted we learned that Maher’s thesis establishes the threshold in substantially greater generality, for invertible polynomials, via the (non)triviality of the primary wall-crossing group [Mah24, Proposition 1.0.5]; we keep the present treatment for its elementary Fermat-case proof, the explicit first-wall census, and the Hamiltonian normal form, on which the scattering construction builds directly.

2. THE COUNT OF CRITICAL GRAPHS

We recall only what is needed and refer to [GKT22] for all definitions. Fix $W = x^r + y^s$. A genus-zero smooth (r, s) -disk carries interior marked points with twists (a_i, b_i) , $0 \leq a_i \leq r-1$,

$0 \leq b_i \leq s-1$, and a descendent vector $\mathbf{d} = (d_i) \in \mathbb{Z}_{\geq 0}$. The interior insertions contributing to the primary potential lie in the Neveu–Schwarz range $0 \leq a_i \leq r-2$, $0 \leq b_i \leq s-2$ [GKT22, eqn. (0.5)].

Given a set J of interior markings with twists $(a_i, b_i)_{i \in J}$ and descendents $\mathbf{d} = (d_i)_{i \in J}$, define, following [GKT22, Notation 2.29],

$$(2) \quad \sum_{i \in J} a_i = r(J) + \ell_1 r, \quad \sum_{i \in J} b_i = s(J) + \ell_2 s,$$

where $r(J) \in \{0, \dots, r-1\}$ and $s(J) \in \{0, \dots, s-1\}$ are the residues, so that

$$(3) \quad \ell_1 = \left\lfloor \frac{1}{r} \sum_{i \in J} a_i \right\rfloor, \quad \ell_2 = \left\lfloor \frac{1}{s} \sum_{i \in J} b_i \right\rfloor.$$

The classification of balanced and critical graphs with interior markings exactly J is [GKT22, Notation 2.38]: setting

$$(4) \quad N(J, \mathbf{d}) = \ell_1 + \ell_2 - |J| + 1 + \sum_{i \in J} d_i,$$

one has $|\text{Balanced}(J, \mathbf{d})| = N(J, \mathbf{d}) + 1$ and

$$(5) \quad |\text{Crit}(J, \mathbf{d})| = \max(N(J, \mathbf{d}), 0).$$

In particular a critical graph with interior markings J and descendent vector $\mathbf{d}$ exists if and only if $N(J, \mathbf{d}) \geq 1$. (Throughout, $J \neq \emptyset$: the boundary-only case $J = \emptyset$, where $N = 1$, comprises the two balanced, non-critical graphs of [GKT22, Example 3.4], and (5) is not asserted there.)

The reduction from invariants to graph counts is the following consequence of [GKT22], which we isolate for use in both directions of Theorem 1.1.

Proposition 2.1. *Fix a finite marking set $I \subseteq \mathbb{N}$ and work over the coefficient ring $A_{I, \text{sym}} = \mathbb{Q}[t_{a,b,d}]/\text{Ideal}(I)$ of [GKT22] (the ideal caps each twist (a,b) at the number of labels of I carrying it). Let $\mathbf{d} = \mathbf{0}$.*

- (a) *If no critical graph with descendent vector $\mathbf{0}$ has interior markings contained in I , then $G_{A_{I, \text{sym}}}^{r,s}$ fixes every primary open FJRW invariant; the primary invariants are canonical.*
- (b) *If some critical graph with descendent vector $\mathbf{0}$ has interior markings contained in I , then two systems of symmetric canonical open FJRW invariants over I differ in a primary invariant.*

Proof. By [GKT22, Definition 4.22] the wall-crossing Lie algebra $\mathfrak{g}_{A_{I, \text{sym}}}^{r,s}$ is the direct sum, over all critical graphs Λ with markings in I , of the one-dimensional spaces $\mathbb{Q} \cdot v_\Lambda$, where for a critical graph with interior markings J , descendents $\mathbf{d}$, and boundary degrees (k_1, k_2) ,

$$(6) \quad v_\Lambda = u_{J, \mathbf{d}} x^{k_1} y^{k_2} ((k_2 + 1)x \partial_x - (k_1 + 1)y \partial_y), \quad u_{J, \mathbf{d}} = \prod_{i \in J} t_{a_i, b_i, d_i}.$$

Here (k_1, k_2) is the index of [GKT22, Definition 4.22]: the critical graph has k_1+1 boundary marked points of type σ_1 and k_2+1 of type σ_2 [GKT22, eqn. (2.31), Proposition 2.37], and the index satisfies the critical condition $sk_1 + rk_2 = w(J, \mathbf{d})$ of Definition 1.2, matching the constraints defining $\mathfrak{g}^{r,s}$; the case $(k_1, k_2) = (0, 0)$ is the marginal torus direction of Definition 1.2. The group is $G_{A_{I, \text{sym}}}^{r,s} = \exp(\mathfrak{g}_{A_{I, \text{sym}}}^{r,s})$ and acts on the set $\Upsilon_{r,s}^{\text{sym}}(I)$ of chamber indices ([GKT22, Definition 3.46]) by $g(W^\nu) = W^{g(\nu)}$ [GKT22, Theorem 4.26].

For (a): if there is no $\mathbf{0}$ -critical graph, every generator (6) has some $d_i > 0$, hence $u_{J,\mathbf{d}}$ is divisible by a variable $t_{a,b,d}$ with $d > 0$. Acting on a potential $W^\nu = x^r + y^s + \sum_{0 \leq a \leq r-2, 0 \leq b \leq s-2} t_{a,b,0} x^a y^b + \dots$ [GKT22, eqn. (0.5)], such a generator raises the total degree in the positive-descendent variables and therefore leaves unchanged the coefficients of the purely primary monomials $u_{J',\mathbf{0}} x^\alpha y^\beta$. Since each primary open invariant $\nu_{\Gamma,\mathbf{0}}$, the coefficient attached to a graph Γ in [GKT22, eqn. (0.4)], is exactly such a coefficient, it is fixed by every generator, hence by $G_{A_{I,\text{sym}}}^{r,s}$.

For (b): a $\mathbf{0}$ -critical graph Λ furnishes a nonzero generator v_Λ with $u_{J,\mathbf{0}} = \prod_{i \in J} t_{a_i,b_i,0}$ a monomial in primary variables only. Fix any system $\nu \in \Upsilon_{r,s}^{\text{sym}}(I)$, a chamber index in the sense of [GKT22, Definition 3.46]; the set is nonempty by [GKT22, Theorem 0.1]. By faithfulness [GKT22, Theorem 4.26, Step 2], $\exp(v_\Lambda)$ does not fix ν ; and to first order, $\exp(v_\Lambda)(W^\nu) - W^\nu = v_\Lambda(x^r + y^s) + \dots$ has the term

$$u_{J,\mathbf{0}} (r(k_2 + 1) x^{k_1+r} y^{k_2} - s(k_1 + 1) x^{k_1} y^{k_2+s}),$$

a primary monomial with a nonzero coefficient. Thus ν and $\exp(v_\Lambda) \cdot \nu$ are two systems of symmetric canonical invariants (both in the single orbit $\Upsilon_{r,s}^{\text{sym}}(I)$, by transitivity) that differ in a primary invariant. $\square$

3. PROOF OF THE CENTRAL-CHARGE CRITERION

The equivalences (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) are immediate: (1) gives $\hat{c}_W < 1 \Leftrightarrow \frac{2}{r} + \frac{2}{s} > 1$, and $\frac{1}{r} + \frac{1}{s} > \frac{1}{2}$ characterizes the two-variable Fermat simple singularities, whose only types are $A_{n-1} = (n, 2)$, $D_4 = (3, 3)$, $E_6 = (3, 4)$ and $E_8 = (3, 5)$ (there is no Fermat E_7 , as $E_7 = x^3 + xy^3$ is not of the form $x^r + y^s$; likewise $D_n = x^{n-1} + xy^2$ is Fermat only for $n = 4$, where $x^3 + y^3 \cong D_4$). It remains to prove (i) $\Leftrightarrow$ (iv), for which, by Proposition 2.1, it suffices to decide the existence of a primary ($\mathbf{d} = \mathbf{0}$) critical graph. The whole matter rests on one inequality.

Lemma 3.1. *Let J be a set of $m = |J|$ interior markings with Neveu–Schwarz twists, $0 \leq a_i \leq r - 2$ and $0 \leq b_i \leq s - 2$. Then, with ℓ_1, ℓ_2 as in (3),*

$$\ell_1 + \ell_2 \leq m \hat{c}_W.$$

Consequently the primary count (4) obeys $N(J, \mathbf{0}) \leq 1 - m(1 - \hat{c}_W)$.

Proof. Dropping the floor functions and using the twist bounds,

$$\ell_1 + \ell_2 = \left\lfloor \frac{1}{r} \sum a_i \right\rfloor + \left\lfloor \frac{1}{s} \sum b_i \right\rfloor \leq \frac{\sum a_i}{r} + \frac{\sum b_i}{s} \leq \frac{m(r-2)}{r} + \frac{m(s-2)}{s} = m \left(2 - \frac{2}{r} - \frac{2}{s} \right) = m \hat{c}_W.$$

The second claim is $N(J, \mathbf{0}) = \ell_1 + \ell_2 - m + 1 \leq m \hat{c}_W - m + 1$. $\square$

Proof of (i) $\Rightarrow$ (iv). Assume $\hat{c}_W < 1$. For any J with $m = |J| \geq 1$ interior markings, Lemma 3.1 gives $\ell_1 + \ell_2 \leq m \hat{c}_W < m$; as $\ell_1 + \ell_2$ is an integer strictly below the integer m , we get $\ell_1 + \ell_2 \leq m - 1$, i.e. $N(J, \mathbf{0}) \leq 0$. By (5) there is no primary critical graph, for any J and any finite I . Proposition 2.1(a) then says the primary invariants are canonical for every I . (The case $m = 0$ is vacuous: there is no interior marking to twist, and $N = 1$ corresponds to the two boundary-only balanced graphs of [GKT22, Example 3.4], which are not critical.) $\square$

Proof of $\neg(i) \Rightarrow \neg(iv)$. Assume $\hat{c}_W \geq 1$, i.e. $\frac{2}{r} + \frac{2}{s} \leq 1$; in particular $r, s \geq 3$ and not both equal to 3, so $r, s \geq 3$ with $\max(r, s) \geq 4$, and we may take all interior twists maximal, $(a_i, b_i) = (r - 2, s - 2)$. With m such markings,

(7)

$$N = \left\lfloor \frac{m(r-2)}{r} \right\rfloor + \left\lfloor \frac{m(s-2)}{s} \right\rfloor - m + 1 = \left(m - \left\lceil \frac{2m}{r} \right\rceil \right) + \left(m - \left\lceil \frac{2m}{s} \right\rceil \right) - m + 1 = m - \left\lceil \frac{2m}{r} \right\rceil - \left\lceil \frac{2m}{s} \right\rceil + 1,$$

using $\lfloor m(r-2)/r \rfloor = \lfloor m - 2m/r \rfloor = m - \lceil 2m/r \rceil$. We exhibit m with $N \geq 1$, i.e. $\lceil 2m/r \rceil + \lceil 2m/s \rceil \leq m$.

If $\frac{2}{r} + \frac{2}{s} < 1$ (the hyperbolic regime $\hat{c}_W > 1$), then $\lceil 2m/r \rceil + \lceil 2m/s \rceil \leq \frac{2m}{r} + \frac{2m}{s} + 2 = m(\frac{2}{r} + \frac{2}{s}) + 2$, which is $\leq m$ once $m(1 - \frac{2}{r} - \frac{2}{s}) \geq 2$; such m exists since $1 - \frac{2}{r} - \frac{2}{s} > 0$.

If $\frac{2}{r} + \frac{2}{s} = 1$ (the parabolic regime $\hat{c}_W = 1$: the pairs $(4, 4)$, $(3, 6)$, $(6, 3)$), choose m with $r \mid 2m$ and $s \mid 2m$, e.g. $m = \text{lcm}(r, s)/2$, an integer for all three parabolic pairs; then $\lceil 2m/r \rceil + \lceil 2m/s \rceil = \frac{2m}{r} + \frac{2m}{s} = m$, so $N = 1$.

In either case a primary critical graph exists once I contains m interior labels of maximal twist, which is possible because each twist is realized infinitely often in the marking data [GKT22, §2.4]. By Proposition 2.1(b) the primary invariants over such an I depend on the boundary conditions. This proves $\neg(\text{iv})$, and with it Theorem 1.1. $\square$

Remark 3.2 (On Remark 0.9 of [GKT22]). [GKT22, Remark 0.9] states that the pairs $(r, s) \in \{(n, 2), (2, n), (3, 3), (3, 4), (4, 3)\}$ give simple singularities “ A_{n-1} , E_6 , and E_8 ”. Two adjustments reconcile the list with the naming. First, $(3, 3)$ appears in the list but D_4 is not named; $x^3 + y^3$ is the D_4 singularity (three distinct lines, Milnor number 4). Second, E_8 is named but its pairs $(3, 5)$, $(5, 3)$ are absent from the list; these are $x^3 + y^5$, Milnor number 8. The corrected wall-crossing-free set is

$$\{(n, 2), (2, n) : n \geq 2\} \cup \{(3, 3), (3, 4), (4, 3), (3, 5), (5, 3)\},$$

of types A_{n-1}, D_4, E_6, E_8 , exactly as in Theorem 1.1(iii). The theorem shows this list is not merely a set of examples but the complete answer to the prediction.

4. THE FIRST WALL AND THE DESCENDENT WALL

Proof of Theorem 1.3. The simple case $m_* = \infty$ is Theorem 1.1. Assume $\hat{c}_W \geq 1$ and use (7), which gives the maximal N over size- m primary multisets, since N is nondecreasing in the twists and maximal at $(r-2, s-2)$. Thus $m_*(r, s)$ is the least m with $\lceil 2m/r \rceil + \lceil 2m/s \rceil \leq m$.

$\min(r, s) \geq 4$. For $m = 1$, $\lceil 2/r \rceil + \lceil 2/s \rceil = 1 + 1 = 2 > 1$ (each $2/r, 2/s \leq \frac{1}{2}$). For $m = 2$ and $r, s \geq 4$, $\lceil 4/r \rceil + \lceil 4/s \rceil = 1 + 1 = 2 = m$, so $N = 1$ and $m_* = 2$.

$\min(r, s) = 3$, say $r = 3$, with $\max = s \geq 6$. Here $a_i \leq 1$, so $\ell_1 = \lfloor m/3 \rfloor$. For $m = 2$, $\lceil 4/3 \rceil + \lceil 4/s \rceil = 2 + 1 = 3 > 2$. For $m = 3$, $\lceil 6/3 \rceil + \lceil 6/s \rceil = 2 + 1 = 3 = m$ (using $s \geq 6$), so $N = 1$ and $m_* = 3$. For $3 \leq s \leq 5$ the pair is simple and $m_* = \infty$, already covered.

Finally $\min(r, s) = 2$ forces the A -type simple case, $m_* = \infty$. This exhausts all (r, s) . For $x^4 + y^4$, the witnessing multiset is two insertions of twist $(r-2, s-2) = (2, 2)$, giving $N = 1$: the example of [GKT26, §6]. $\square$

Proof of Proposition 1.4. Take $|J| = 1$ with any twist. Then $\ell_1 = \ell_2 = 0$ and, by (4), $N(J, \mathbf{d}) = 0 + 0 - 1 + 1 + d = d$, where d is the descendent degree of the single insertion. Hence $d = 1$ gives $N = 1$: exactly one critical graph, for every (r, s) . Proposition 2.1(b) is stated for $\mathbf{d} = \mathbf{0}$, but its proof uses only that the generator is nonzero and that the group acts faithfully: applied to the descendent-1 generator, the same first-order computation produces a monomial divisible by $t_{a,b,1}$ with nonzero coefficient, so two systems of invariants differ in a descendent invariant. Hence the descendent theory wall-crosses in every case. $\square$

Remark 4.1. The trichotomy of Theorem 1.3 shows the $x^4 + y^4$ example of [GKT26] is not isolated: two interior insertions already force a wall for every (r, s) with $r, s \geq 4$, and three suffice whenever $\min(r, s) = 3 < \frac{1}{2} \max(r, s)$. Combined with Proposition 1.4, the qualitative picture is complete: the descendent-free theory is canonical exactly on the ADE locus, while switching on a single descendent breaks canonicity everywhere.

5. THE WALL GENERATORS ARE HAMILTONIAN

We now prove Proposition 1.5. Throughout, $\omega = dx \wedge dy$, and for a bivariate polynomial H the Hamiltonian vector field is $X_H := (\partial_y H) \partial_x - (\partial_x H) \partial_y$, so that $\iota_{X_H} \omega = dH$. In the rank-two case the wall-crossing Lie algebra $\mathfrak{g}^{r,s}$ of [GKT22] is a sum of one-dimensional spaces spanned by

$$(8) \quad X_{a,b} = x^a y^b ((b+1)x \partial_x - (a+1)y \partial_y) = (b+1)x^{a+1} y^b \partial_x - (a+1)x^a y^{b+1} \partial_y,$$

one for each critical graph, the exponents $(a, b) = (k_1, k_2)$ being its [GKT22, Definition 4.22] index: the graph has k_1+1 and k_2+1 boundary marked points of the two types, so the Hamiltonian $H_{k_1, k_2} = x^{k_1+1} y^{k_2+1}$ of Lemma 5.1 below is its boundary monomial.

Lemma 5.1. *For all $a, b \geq 0$, $X_{a,b} = X_{H_{a,b}}$ with $H_{a,b} = x^{a+1} y^{b+1}$. In particular $\operatorname{div} X_{a,b} = 0$, so $X_{a,b}$ preserves ω .*

Proof. $\partial_y H_{a,b} = (b+1)x^{a+1} y^b$ and $\partial_x H_{a,b} = (a+1)x^a y^{b+1}$, so $X_{H_{a,b}} = (b+1)x^{a+1} y^b \partial_x - (a+1)x^a y^{b+1} \partial_y$, which is (8). A Hamiltonian vector field is divergence-free: $\operatorname{div} X_{a,b} = \partial_x \partial_y H_{a,b} - \partial_y \partial_x H_{a,b} = 0$, recovering the volume-preservation of [GKT22, p. 85]. $\square$

Lemma 5.2. *For all $a, b, c, d \geq 0$,*

$$(9) \quad [X_{a,b}, X_{c,d}] = ((b+1)(c+1) - (a+1)(d+1)) X_{a+c, b+d}.$$

Equivalently, with the Poisson bracket $\{H, K\} := X_H(K)$ induced by ω ,

$$\{H_{a,b}, H_{c,d}\} = ((b+1)(c+1) - (a+1)(d+1)) H_{a+c, b+d}.$$

Proof. Write $X_{a,b} = f \partial_x + g \partial_y$ with $f = (b+1)x^{a+1} y^b$, $g = -(a+1)x^a y^{b+1}$, and $X_{c,d} = F \partial_x + G \partial_y$ similarly. A direct computation of $[X_{a,b}, X_{c,d}] = (X_{a,b}F - X_{c,d}f) \partial_x + (X_{a,b}G - X_{c,d}g) \partial_y$ gives, for the ∂_x -coefficient,

$$X_{a,b}F - X_{c,d}f = (b+d+1)[(b+1)(c+1) - (a+1)(d+1)] x^{a+c+1} y^{b+d},$$

which is $((b+1)(c+1) - (a+1)(d+1))$ times the ∂_x -coefficient $(b+d+1)x^{a+c+1} y^{b+d}$ of $X_{a+c, b+d}$; the ∂_y -coefficient matches identically. For the Poisson form, $\{H_{a,b}, H_{c,d}\} = X_{a,b}(H_{c,d}) = (b+1)(c+1)x^{a+c+1} y^{b+d+1} - (a+1)(d+1)x^{a+c+1} y^{b+d+1}$, and $X_{H_{a+c, b+d}} = X_{a+c, b+d}$ by Lemma 5.1 shows the two identities agree. $\square$

The determinant $\langle (a, b), (c, d) \rangle = (b+1)(c+1) - (a+1)(d+1)$ appearing in (9) is the structure constant of the tropical vertex Lie algebra of Kontsevich–Soibelman [KS06]; see [Gro10]. Assembling Lemmas 5.1 and 5.2 proves Proposition 1.5: $H_{a,b} \mapsto X_{a,b}$ is a Lie-algebra isomorphism from the Poisson algebra on $\{x^p y^q : p, q \geq 1\}$ onto its image, restricting to an isomorphism onto $\mathfrak{g}^{r,s}$ once one keeps only the Hamiltonians H_Λ of critical graphs. The grading is immediate from the critical condition.

Proposition 5.3. *Let Λ be a critical graph with interior markings J , descendants $\mathbf{d}$, and index (k_1, k_2) , and set $m(\Lambda) = m(J, \mathbf{d}) = rs + \sum_{i \in J} (sa_i + rb_i + rs(d_i - 1))$ [GKT22, Notation 2.29]. Then the Hamiltonian $H_\Lambda = x^{k_1+1} y^{k_2+1}$ has (s, r) -weighted degree*

$$s(k_1 + 1) + r(k_2 + 1) = m(\Lambda) + r + s - rs.$$

Thus $\mathfrak{g}^{r,s}$ is graded by the (s, r) -weighted degree of the wall Hamiltonian, and the degree records the mass of the critical graph (equivalently its Euler weight $w(\Lambda) = m(\Lambda) - rs$ of Definition 1.2).

Proof. The critical condition is $sk_1 + rk_2 = m(\Lambda) - rs$ [GKT22, Proposition 2.37]; add $r + s$. $\square$

Remark 5.4 (Relation to the tropical vertex algebra). The tropical vertex Lie algebra is the Hamiltonian algebra of the *torus* form $d \log x \wedge d \log y$, for which $\{x^u, x^v\} = \langle u, v \rangle x^{u+v}$ with $\langle u, v \rangle$ the determinant, so that Hamiltonians add exponents. Proposition 1.5 uses instead the *affine-plane* form $dx \wedge dy = xy (d \log x \wedge d \log y)$, under which $\{x^u, x^v\} = \langle u, v \rangle x^{u+v-(1,1)}$. The two Poisson structures, hence the two Lie algebras, are intertwined by the exponent shift $u \mapsto u - (1, 1)$ induced by this factor xy – the same change of primitive form that makes $dx \wedge dy$ the natural volume in [GKT22, Theorem 0.3]. So $\mathfrak{g}^{r,s}$ is precisely an affine-plane avatar of a tropical-vertex Lie algebra, cut out by the critical Hamiltonians, which makes explicit the analogy that [GKT22, Remarks 0.8 and 4.19] draw “in spirit” with the Kontsevich–Soibelman group. The genuine scattering diagram whose walls are the critical graphs is constructed in [Wue26]: there the set of invariant systems is presented as the chamber set of a consistent scattering structure, a canonical diagram with computable wall functions is extracted, and the potentials are shown to be broken-line transports of a canonical seed system. The tropical open FJRW theory called for in [GKT26, §6] is posed there through tropical moduli, a priori a distinct route to the same tropical picture.

6. THE FIRST WALL AT THE PARABOLIC BOUNDARY

The threshold $\hat{c}_W = 1$ is not only where wall-crossing begins across the family; the walls it produces have a uniform shape. By the *first wall* of (r, s) we mean the set of critical graphs over interior multisets of the minimal size $m_*(r, s)$. Combining the count (5) with the Hamiltonian dictionary of §5 gives a complete census of the first wall for each (r, s) ; at the parabolic boundary it collapses to a single wall.

Proposition 6.1 (Parabolic first wall). *For any interior multiset J with $N(J, \mathbf{0}) = 1$, the index of its unique critical graph is $(k_1, k_2) = (r(J), s(J))$, of Euler weight $sr(J) + rs(J)$. For the three parabolic pairs, $(r, s) \in \{(4, 4), (3, 6), (6, 3)\}$ with $\hat{c}_W = 1$, the first wall consists of exactly one critical graph, its multiset forced to be the maximal-twist one: two copies of $(2, 2)$ for $(4, 4)$, three copies of $(1, 4)$ for $(3, 6)$, three copies of $(4, 1)$ for $(6, 3)$. Its residues vanish, so its index is $(0, 0)$, its weight is zero, and its direction is the marginal torus direction $x\partial_x - y\partial_y$ of Definition 1.2, with Hamiltonian xy . By contrast, for the hyperbolic pairs of Table 1 the first wall comprises several critical graphs, both genuine positive-weight walls and marginal torus directions.*

Proof. If $N(J, \mathbf{0}) = 1$ there is a unique critical graph and, by [GKT22, Notation 2.38, eqn. (2.31)] read in the indexing of (6), its index is $(k_1, k_2) = (r(J), s(J))$; the weight statement is the critical condition of Definition 1.2. For each parabolic pair the minimal critical multiset is uniquely the maximal-twist one: at $m = m_*$ the inequality $\ell_1 + \ell_2 \geq m$ is attained only with equality and only at $(a_i, b_i) = (r - 2, s - 2)$ (for $(4, 4)$, $\ell_1, \ell_2 \leq 1$ force $a_i = b_i = 2$; for $(3, 6)$, $\ell_1 \leq 1, \ell_2 \leq 2$ force $a_i = 1, b_i = 4$). There the residues vanish, so $(k_1, k_2) = (0, 0)$: the torus direction, with Hamiltonian $H_{0,0} = xy$ (Lemma 5.1). The hyperbolic data is the direct enumeration recorded in Table 1. $\square$

The parabolic pairs are the two-variable Fermat *simply-elliptic* singularities $(x^4 + y^4 = \tilde{E}_7, x^3 + y^6 = \tilde{E}_8)$, for which the primitive form ceases to be $dx \wedge dy$ and becomes a period of an elliptic curve; the marginal torus direction is where that transition first registers in the open theory: the elliptic wall-crossing group of [Mah24, §5.2.2] is exactly this torus factor,

(r, s)	$\hat{c}_W$	regime	m_*	graphs	marginal	wall Hamiltonians
(4, 4)	1	parabolic	2	1	1	— (torus xy only)
(3, 6)	1	parabolic	3	1	1	— (torus xy only)
(6, 3)	1	parabolic	3	1	1	— (torus xy only)
(4, 5)	11/10	hyperbolic	2	2	1	xy^2
(3, 7)	22/21	hyperbolic	3	2	1	xy^2
(4, 6)	7/6	hyperbolic	2	4	2	xy^2, xy^3
(5, 5)	6/5	hyperbolic	2	5	2	xy^2, x^2y, x^2y^2

TABLE 1. The first wall of $x^r + y^s$: the number of critical graphs at $m = m_*(r, s)$, how many are marginal (index $(0, 0)$, the torus direction with Hamiltonian xy), and the boundary monomials $x^{k_1+1}y^{k_2+1}$ Hamiltonian to the genuine wall generators. At the parabolic boundary the first wall is purely marginal; genuine walls appear only in the hyperbolic regime. Computed exactly; see §7.

and the explicit jumping of the $x^4 + y^4$ invariants in [GKT26, Example 4.9] is its action. Genuine (pro-nilpotent) walls appear precisely in the hyperbolic regime. The first *symmetric* hyperbolic Fermat pair is $x^5 + y^5$, with boundary monomials xy^2, x^2y, x^2y^2 in Table 1; its wall functions and scattering diagram are computed in [Wue26].

7. VERIFICATION

The combinatorial and algebraic claims are verified exactly by the accompanying program `src/census.py`, published with this note and the companion paper in the repository of [Wue26], which uses only integer, `fractions.Fraction`, and exact symbolic arithmetic. It transcribes (4) from [GKT22, Notation 2.38] and checks the following.

- Over the grid $2 \leq r, s \leq 40$: the threefold identity of Theorem 1.1(i)–(iii), and that the pairs without a primary critical graph are exactly the ADE Fermat list, with $D_4 = (3, 3)$ and $E_8 = (3, 5), (5, 3)$ present (Remark 3.2). For $\hat{c}_W < 1$ the negative is proved rather than merely searched, Lemma 3.1 bounding the search by $m \geq 1/(1 - \hat{c}_W)$, beyond which $N \leq 0$ is forced.
- The trichotomy for $m_*(r, s)$ of Theorem 1.3.
- The first-wall census of Table 1 with its marginal/genuine split, each positive-weight index cross-checked against the independent implementation of [GKT22, Definition 4.22] in `src/gkt_algebra.py` of [Wue26], and the parabolic statement of Proposition 6.1.
- The universal descendent wall of Proposition 1.4, including its refinement (torus direction at the zero twist, genuine wall of index (a, b) otherwise).
- By exact polynomial arithmetic on vector fields: the Hamiltonian identity of Lemma 5.1, the divergence-freeness, and the bracket (9) of Lemma 5.2 over $0 \leq a, b, c, d \leq 4$.

CONCLUDING REMARKS AND OPEN DIRECTIONS

Of three natural continuations, the first, promoting the Hamiltonian identification to a genuine scattering diagram whose walls are the critical graphs, is carried out in the companion paper [Wue26]: the set of open FJRW invariant systems is presented there as the chamber set of a consistent scattering structure over $\mathfrak{g}^{r,s}$, a canonical diagram $\mathfrak{D}^{r,s}$ with computable wall functions is constructed from unique extreme chamber indices, the open topological recursion

appears as the direction transverse to the walls, and the potential family is shown to be the broken-line transport of a canonical seed system. The $(1, 1)$ -shift of Remark 5.4 is precisely the mechanism that forbids backscattering of broken lines there. Two directions remain open.

- (1) *Explicit ADE potentials.* On the wall-crossing-free locus the primary potential is unambiguous; computing it in closed form for D_4 , E_6 , E_8 (matching the D -series potentials of [BB21] and extending them to the E -cases) and relating the coefficients to Coxeter/root-system data is a concrete computational program.
- (2) *The parabolic boundary.* Proposition 6.1 pins the first wall of the parabolic pairs to the marginal torus direction. What remains is the analytic side: these are the simply-elliptic singularities $\tilde{E}_7, \tilde{E}_8$, whose primitive form is a period of an elliptic curve, so the flat structure is governed by modular forms rather than polynomials [Mah24, §4.2.2]. Identifying the modular period map on the torus factor, beyond the $x^4 + y^4$ jumping of [GKT26, Example 4.9], is the first genuinely transcendental case of the theory.

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